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BroadLineRegions.jl: A fast and flexible toolkit for modeling the broad-line region (BLR) in Julia.

Kirk Long [1*]

1 JILA, University of Colorado Boulder, USA **2** Kavli Institute for Astronomy and Astrophysics, Peking University, China * These authors contributed equally.

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Summary

Quasars and active galactic nuclei (AGN) show a remarkable degree of line-broadening, an observational hallmark whose origin is assumed to be due to the fast motions of line-emitting gas near the central supermassive black hole (Cackett et al., 2021). The region from which this line emission originates is aptly named the broad-line region (BLR). Measuring and constraining properties of the BLR thus provides one of the only opportunities to directly constrain the masses of supermassive black holes outside our local universe as well as to better understand the environments directly surrounding them. For example, assuming the gas is virialized we in principle need to measure just three quantities in the BLR to get the black hole mass (Cackett et al., 2021):

$$M_{\rm BH} = f\frac{R_{\rm BLR}\left(\Delta V\right)^2}{G}$$

Here ΔV is the characteristic velocity of the gas, which is often approximated from the width of the broad-line and can be estimated from even just a single spectrum of the source. $R_{\rm BLR}$ is the characteristic distance from the supermassive black hole to the BLR, which has historically been constrained via reverberation mapping (RM) (Blandford & McKee, 1982; Cackett et al., 2021) and more recently has been constrained with interferometric observations with GRAVITY (Abuter et al., 2024; GRAVITY Collaboration et al., 2019). f is the so-called "virial factor", which encodes extra information about the geometry and kinematics of the BLR, and G is Newton's gravitational constant. While the black hole mass can be very roughly obtained from simple estimates of each of these quantities, to accurately measure the masses of supermassive black holes across cosmic time we must interpret data through a model of the BLR. There are thus both significant measurement and model-dependent uncertainties encoded in ΔV, $R_{\rm BLR}$ and f. Sources with supermassive black hole masses measured from their broad-line regions underlie the measurements of many further reported black hole measurements, as they are used as calibrators in scaling relations (GRAVITY Collaboration et al., 2024).

Beyond measuring black hole masses, if we want to understand the environment surrounding supermassive black holes we must characterize the kinematics and geometry of the BLR. There is a wealth of exciting new data from velocity-resolved reverberation mapping campaigns as well as interferometric observations with GRAVITY that present new opportunities to unravel the fundamental nature of the BLR, but to do so requires a degree of modeling to explain the data.

Existing modeling techniques work well to explain individual data products, but self-consistently explaining all of the data products from reverberation and spectroastrometic observations will likely require more flexible dynamical models than those that have been commonly used in the literature (Long & Dexter, 2025). `BroadLineRegions.jl` was designed to fulfill this need, and

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at the highest level allows users to flexibly define more complicated dynamical BLR models by combining multiple sub-models together with simple syntax. It integrates with existing popular classes of dynamical BLR models, as well as enabling support for fully custom models. After creating a model with `BroadLineRegions.jl` one can then generate various observables to compare to observational data products from reverberation and spectroastrometric campaigns.

Statement of need

`BroadLineRegions.jl` enables fast and (more importantly) flexible modeling of the BLR in Julia, allowing for the quick creation of novel model BLRs for comparison to data or for making theoretical predictions. There are many possible models of the BLR, and in reality the BLR is likely more complicated than any single component model can describe (Long & Dexter, 2025). `BroadLineRegions.jl` enables researchers to test different multi-component BLR possibilities quickly by allowing for both new and existing BLR component models to be combined with simple syntax. This flexibility allows researchers to easily test their own bespoke models against and in concert with other more standard pictures of the BLR. To most accurately measure the masses of supermassive black holes as well as to better understand the fundamental nature of the BLR we must understand what classes of dynamical models best fit observations. Additionally, many existing BLR modeling tools are closed source and thus `BroadLineRegions.jl`'s open-source philosophy will enable anyone to use it as a tool to uncover the true nature of the gas surrounding supermassive black holes. This further allows for researchers to independently contribute their models and ideas to this tool in an open and transparent way to enable new advances for everyone in our understanding of the true nature of the BLR and how to most accurately recover supermassive black hole masses. Finally, `BroadLineRegions.jl` allows for easier interfacing with GRAVITY and RM data products simultaneously, whereas most other BLR modeling codes are focused solely on the reverberation problem.

State of the field

Currently researchers working to understand the BLR through modeling take one of two approaches: either they use simple, largely kinematic models to quickly fit the observations (Chiang & Murray, 1996; Li & Wang, 2023; Pancoast et al., 2011, 2014; Rosborough et al., 2024), or they attempt more in-depth and physically realistic simulations of the fluids and radiative transfer to try to produce similar results to observations from the ground up (Mangham et al., 2019; Waters et al., 2016). While the second approach is admirable, it is currently computationally infeasible to apply such detailed modeling to actually fit the data observed in sources, thus this code follows the first approach — being largely a kinematic, simple modeling tool designed to flexibly fit and match data with a more limited set of physics.

Currently the most popular kinematic modeling code used to fit the BLR is the closed-source `CARAMEL` code as described in Pancoast et al. (2011) and Pancoast et al. (2014). While a powerful tool to flexibly fit BLR line profiles, as pointed out in (Mangham et al., 2019) and Long & Dexter (2025) this model of the BLR cannot fully explain more recent velocity-resolved reverberation mapping measurements. Other modeling codes include BRAINS (Li & Wang, 2023), which is open-source, fast, and capable of modeling both spectroastrometric and reverberation mapping data products, but is somewhat rigid in both its modeling scope and input requirements and also has not yet demonstrated that is capable of self-consistently reproducing all of the available reverberation mapping data products in non-standard BLRs. More recently the closed-source BELMAC (Rosborough et al., 2024) code has also been developed, which is a new kinematic/geometric model of the BLR that attempts to combine the best parts of `CARAMEL` style modeling with more realistic physics, but it is again a more rigid modeling framework than what we present here and has not yet been applied to interferometric data nor yet demonstrated that it is capable of self-consistently matching all of the possible

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reverberation mapping data products in more complicated sources. The rigidity in all of these codes is in their preferred geometries, kinematics, and expression of parameters, which while well-motivated to explain certain aspects of the observations can be difficult to modify in the ways that are required to self-consistently explain all of the data. These facts motivated the development of `BroadLineRegions.jl` to fill the need to flexibly model the BLR for all kinds of data products, enabling the combination and testing of many different classes of geometric and kinematic models against each other in many different contexts.

Software design

`BroadLineRegions.jl` was built from the ground-up to be highly modular and flexible, such that end-users can replace certain parts/steps with custom/novel alternatives if they wish to. This goal is motivated by the scientific objective that at time of publication no existing modeling code has demonstrated the ability to fully explain all of the reverberation and GRAVITY data products self-consistently (Long & Dexter, 2025; Mangham et al., 2019), thus clearly more flexible models are needed. `BroadLineRegions.jl` is agnostic on exactly *what* the "true" model of the BLR is, we only wish to provide the required tools to build the kind of multi-component models that will be needed to self-consistently model all data products from the BLR.

By default we supply constructors for two popular classes of kinematic models applied to the BLR in the literature: "cloud" and "disk-wind" style BLR models. The "cloud" model is similar in implementation to the popular `CARAMEL` BLR implementation (Pancoast et al., 2011, 2014) and the "disk-wind" model similar in implementation to an idea first described by Chiang and Murray (Chiang & Murray, 1996). While neither of these models are novel by themselves, the design of `BroadLineRegions.jl` is novel in how it enables new ways for them to be implemented and tested. We allow users to easily swap in and out components of these models and how they are parameterized, and we enable them to be flexibly combined to make more complicated multi-component models of the BLR that are not possible with existing frameworks. Of course users are not restricted to using these constructors, and another key strength of `BroadLineRegions.jl` is we make it simple for users to define entirely custom models which can be integrated directly along side the more "common" BLR models discussed above.

For example, if a user wanted to model the BLR as a hybrid model of a disk-wind and cloud setup, this can be accomplished in just three lines of code: one to initialize the cloud model, one to initialize the disk-wind model, and a third to create the combined model, an approach demonstrated in Long & Dexter (2025). This kind of behavior is also possible with fully custom models, i.e. a user can also swap in a fully custom modeled component in the preceding example. Simpler modifications are also possible of course, i.e. suppose an end-user may like the cloud model of the BLR similar to how it has been described in Pancoast et al. (2014) but wishes to alter the intensity from the default setting — this is again easy to accomplish as a result of the modular nature of the code, as they can just pass a custom intensity function to the cloud model constructor and `BroadLineRegions.jl` will call this custom intensity function instead of the default when generating the model.

We use several mutable structs to accomplish this behavior, the most important being the `ring` struct, which represents a ring on a "camera" that represents the observer's view of the BLR. A single `ring` is created by defining the relevant physical quantities (intensity, velocity, distance from the source, etc.) either by hand or through helper functions, which can be user supplied and thus enabling great flexibility for the end-user. A `model` is then a combination of camera `rings`, and the code has been designed such that one can easily combine models together to build up the BLR piece by piece. This mutability of our key data structures enables the easy combination of models and allows for the user to overwrite quantities after model creation. The documentation attempts to make this distinction very clearly to warn users that models are not permanent, but of course this may also introduce some user confusion if they

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accidentally overwrite a variable and expected it to stay constant, and there may be a small performance loss from this mutability design choice. Despite this modest tradeoff, we believe that this approach is required scientifically because this software was motivated by the idea that current BLR modeling codes are too rigid to explain all the data, and thus we designed `BroadLineRegions.jl` to intentionally use this flexibility as a distinct feature advantage.

`BroadLineRegions.jl` can work entirely on the CPU alone but can also generate models and perform calculations using both CUDA and Metal (Apple) devices. It is also scaleable from small machines to larger ones, with all capabilities possible on even small personal machines but with larger machines required to generate many models in fitting data more quickly. Many routines in both model generation and observable calculation are parallelized, both on the CPU and on the GPU, thus the performance of the code improves with additional computational resources in workflows where models are repeatedly created and observed such as in fitting to data. Since Julia is a "just in time" (JIT) compiled language, initial model calls when first loading the package or when changing models are much slower than subsequent model calls.

Research impact statement

`BroadLineRegions.jl` has already been used to better model the BLR and expose the systematic uncertainties in our choice of models (Long et al., 2023, 2025; Long & Dexter, 2025). As demonstrated in Long & Dexter (2025), the BLR is likely more complicated than current existing single-component kinematic models can describe, thus motivating the newly possible approaches that can be taken with this code. To truly understand the environments around supermassive black holes as well as most accurately measure their masses we must have models that are capable of matching all of the data products, thus `BroadLineRegions.jl` has the potential to be a next-generation open-source tool for accomplishing this goal.

AI usage disclosure

No generative AI was used in the preparation of the scientific content of this manuscript. The foundations of the software described in this work were completed before the ubiquity of current LLM coding tools, and for versions prior to v0.2 no LLMs were used in generating code. More recently generative AI coding tools have improved substantially, and in this software we have utilized them to implement a well-scoped performance-oriented refactor of the core raytracing and transfer-function routines (yielding order-of-magnitude speedups over the original brute-force implementation) as well as the addition of GPU acceleration (a CUDA/`KernelAbstractions.jl` backend supporting device-resident models and direct on-device model construction), together with the corresponding documentation. This work was carried out using Anthropic's Claude Code (Opus 4.8 and Fable 5 models) and OpenAI's ChatGPT Codex (GPT-5.5) to test the different model capabilities. In every case the author specified the design and acceptance criteria, directed the implementation, and reviewed and tested all changes against the pre-existing CPU reference implementation to ensure correctness. These tools were used as implementation and optimization aids and not to make scientific or modeling design decisions, which remain entirely the author's. LLMs have been used for coding tasks in this package from v0.2 onwards, thus the split between primarily human generated code vs the LLM refactor/GPU support additions can be easily seen by inspecting the changes in the release of these two versions, which also include descriptive PR messages further detailing all changes.

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